

CYBERSECURITY 60-DAY CHALLENGE

Task 1 – Foundation & Environment Setup

LAB SETUP REPORT

ApexPlanet Internship / Training Program

Student Name: _____

Date: _____

1. Objective

The objective of this task was to build a safe, isolated cybersecurity practice laboratory and become familiar with essential Linux and security tools. The lab consists of Kali Linux as the security testing machine and Metasploitable2 as an intentionally vulnerable target. The target system is connected only to a private Host-Only network.

2. Host Machine and Virtualization

Component	Configuration
Host OS	Windows 11 Home Single Language, 64-bit
Processor	Intel 12th Gen Core i3-1215U
RAM	8 GB
Virtualization	Enabled
VirtualBox	7.2.16
Lab Storage	D:\Cybersecurity-Lab

3. Kali Linux Configuration

Kali Linux 2026.2 (VirtualBox image); 2048 MB RAM; 2 CPUs; Adapter 1 = NAT; Adapter 2 = Host-Only Adapter; eth0 = 10.0.2.15/24; eth1 = 192.168.56.101/24.

Evidence: [INSERT SCREENSHOT: Kali Linux desktop/terminal showing the VM running and IP configuration]

Figure 1. Kali Linux running in the VirtualBox lab.

4. Metasploitable2 Configuration

Metasploitable2 was configured with 512 MB RAM, 1 CPU, and a Host-Only network adapter. Its eth0 address was 192.168.56.102/24. The default training login was msfadmin / msfadmin.

Evidence: [INSERT SCREENSHOT: Metasploitable2 terminal showing ifconfig]

Figure 2. Metasploitable2 running on the isolated Host-Only network.

5. Network Architecture

Internet → NAT → Kali Linux (10.0.2.15)

Kali Linux (192.168.56.101) ↔ Host-Only 192.168.56.0/24 ↔ Metasploitable2 (192.168.56.102)

The vulnerable target was kept on the Host-Only network without a NAT/Internet adapter.

6. Connectivity Verification

```
ping -c 4 192.168.56.102
```

Result: 4 packets transmitted, 4 packets received, 0% packet loss.

[INSERT SCREENSHOT: Kali ping result]

Figure 3. Successful connectivity between Kali and Metasploitable2.

7. Nmap Host Discovery and Service Scan

```
nmap -sn 192.168.56.0/24
```

```
nmap 192.168.56.102
```

Host discovery identified the VirtualBox interfaces, Kali at 192.168.56.101, and Metasploitable2 at 192.168.56.102. The service scan found numerous open services, including FTP 21, SSH 22, Telnet 23, HTTP 80, SMB 139/445, MySQL 3306, PostgreSQL 5432, VNC 5900, IRC 6667, AJP 8009, and others. This is expected for the intentionally vulnerable Metasploitable2 training target.

[INSERT SCREENSHOT: Nmap service scan]

Figure 4. Nmap scan of the Metasploitable2 target.

8. Wireshark Packet Capture

Wireshark was used on Kali's eth1 Host-Only interface. With the display filter **icmp**, the capture showed four Echo Requests from 192.168.56.101 to 192.168.56.102 and four corresponding Echo Replies.

[INSERT SCREENSHOT: Wireshark with icmp filter and request/reply packets]

Figure 5. Wireshark ICMP capture.

9. Netcat TCP Communication Test

```
Metasploitable2: nc -l -p 4444
```

```
Kali: nc -v 192.168.56.102 4444
```

A text message was successfully transmitted from Kali and displayed on Metasploitable2, confirming TCP communication over the isolated lab network.

[INSERT SCREENSHOT: Netcat listener and client/message]

Figure 6. Netcat TCP communication test.

10. Burp Suite HTTP Interception

Burp Suite Community Edition was used with its built-in browser. Accessing <http://192.168.56.102> produced a GET request for '/' in Proxy → HTTP history, verifying HTTP traffic interception.

[INSERT SCREENSHOT: Burp Suite HTTP history]

Figure 7. HTTP request captured in Burp Suite.

11. OpenSSL Cryptography Practical

```
openssl version
```

```
openssl genrsa -out private.key 2048
```

```
openssl rsa -in private.key -pubout -out public.key
```

```
openssl req -x509 -new -key private.key -out certificate.crt -days 365
```

```
openssl x509 -in certificate.crt -text -noout
```

OpenSSL 3.6.2 was verified. A 2048-bit RSA private/public key pair and a self-signed X.509 certificate were generated. The certificate used the Common Name `metasploitable.local` and was inspected successfully.

[INSERT SCREENSHOT: OpenSSL certificate details]

Figure 8. OpenSSL-generated X.509 certificate details.

12. Linux Fundamentals Practiced

Command	Purpose
<code>pwd</code>	Display current directory
<code>ls</code>	List files/directories
<code>cd</code>	Change directory

chmod	Change permissions
chown	Change owner/group
apt	Package management
dpkg	Debian package management
ifconfig	Display/configure interfaces
ping	Test connectivity
netstat	Display network connections
traceroute	Show network path

[INSERT SCREENSHOT: Linux command practice]

Figure 9. Linux fundamentals command practice.

13. Security and Isolation

Metasploitable2 was kept on the Host-Only network and was not given a NAT/Internet adapter. Kali used NAT for Internet access and a separate Host-Only interface for the private lab. This separation reduces the risk of exposing the intentionally vulnerable target outside the training environment.

14. Tools Familiarized

Nmap – host discovery and service enumeration
Wireshark – packet capture and protocol analysis
Netcat – basic TCP communication testing
Burp Suite – HTTP request interception and inspection
OpenSSL – cryptographic keys and X.509 certificates

15. Conclusion

The Foundation & Environment Setup task was completed by creating an isolated VirtualBox cybersecurity lab containing Kali Linux and Metasploitable2. Connectivity was verified, services were enumerated with Nmap, ICMP traffic was analyzed with Wireshark, TCP communication was tested with Netcat, HTTP traffic was inspected with Burp Suite, and basic cryptographic operations were performed with OpenSSL. Linux networking and administration commands were also practiced.

16. Screenshot Checklist

- Kali Linux running
- Metasploitable2 running with ifconfig
- Kali IP configuration
- Successful Kali → Metasploitable2 ping
- Nmap host discovery
- Nmap service scan
- Wireshark ICMP capture
- Netcat connection/message
- Burp Suite HTTP history
- OpenSSL certificate details
- Linux command practice